Supplementary Material for Proposition 26

This supplement proves Proposition 26 on codeword supports of weight $2j+2$ and gives the computer certificates used in Theorem 30. The notation and the projection convention agree with the paper.

S1. Notation and Preliminaries

Let p be an odd prime, and let $A, B \le \mathbb{F}_p^\times$ have orders k, j, respectively. For a point set $\mathcal{P} \subseteq \mathbb{F}_p^2$, write $\pi_\lambda(u,x) = x + \lambda u$. A finite direction λ is even if every fiber of π_λ on $\mathcal{P}$ has even cardinality. The direction ∞ is even if every fixed-u column has even occupancy. Write $\mathcal{E}(\mathcal{P})$ for the set of all even directions, including ∞ when applicable. A point set $\mathcal{P} \subseteq A \times \mathbb{F}_p$ is a codeword support precisely when $B \subseteq \mathcal{E}(\mathcal{P})$.

We recall the even-direction bound of Lemma 9 of the paper.

Lemma 9. *Let p be an odd prime, and let $\mathcal{P} = \{(u_i, x_i) : 1 \le i \le 2h\} \subseteq \mathbb{F}_p^2$ consist of distinct points, where $h \ge 1$. If r distinct finite directions are even for $\mathcal{P}$, then $h \ge r$.*

For $\mathcal{P} = \{(u_i, x_i) : 1 \le i \le 2h\}$, use the Rédei polynomial [1]

$$R_{\mathcal{P}}(z,y) = \prod_{i=1}^{2h} (z - x_i - y u_i).$$

Let $Q_{\mathcal{P}}(z,y) = z^h + \sum_{s=1}^{h} q_s(y) z^{h-s}$ be the unique monic polynomial obtained by matching the coefficients of $z^{2h-1}, \dots, z^h$ in $R_{\mathcal{P}}$ and $Q_{\mathcal{P}}^2$, and put

$$D_{\mathcal{P}} = R_{\mathcal{P}} - Q_{\mathcal{P}}^2 = \sum_{i=0}^{h-1} d_i(y) z^i.$$

The proof of Lemma 9 gives $\deg_y q_s \le s$, $\deg_y d_i \le 2h - i$, and $(y-\lambda)^2 \mid d_i(y)$ for every finite even direction λ and every $0 \le i < h$.

S2. Proof of Proposition 26

Proposition 26. *Let p be an odd prime, and let $A, B \le \mathbb{F}_p^\times$ have orders k, j, respectively, where $3 \le j < k < p-1$. Suppose that a codeword support $\mathcal{P} \subseteq A \times \mathbb{F}_p$ has size $2j+2$. Then the following hold.*

(i) *Some $u \in A$ occurs as the first coordinate of an odd number of points of $\mathcal{P}$.*

(ii) *With the notation of Lemma 9, $D_{\mathcal{P}}(z,y) = (y^j - 1)^2 G_{\mathcal{P}}(z,y)$ for a nonzero polynomial $G_{\mathcal{P}}$ of total degree at most two. Either $\mathcal{P}$ lies on a smooth conic, or $-G_{\mathcal{P}}$ is a square in $\mathbb{F}_p[z,y]$.*

(iii) *If $j \ge 4$, then $\mathcal{P}$ does not lie on a smooth conic. Moreover, there are distinct monic polynomials $P_+(z), P_-(z)$ of degree $j+1$, all of whose roots lie in A, such that $\deg(P_+ - P_-) \le 1$. Roots are counted with multiplicity.*

Proof. Since k is a proper divisor of $p-1$, we have $p \ge 2k+1 > 2j+2$.

Proof of (i). Suppose every column occupancy is even. Then $B \cup \{\infty\} \subseteq \mathcal{E}(\mathcal{P})$. By Lemma 9, at most $j+1$ finite directions are even, so choose a finite $\lambda_0 \notin \mathcal{E}(\mathcal{P})$. Apply $u' = x + \lambda_0 u$, $x' = u$, and denote the transformed support by $\mathcal{P}'$. The induced map of directions is $\lambda \mapsto 1/(\lambda - \lambda_0)$, with $\lambda_0 \mapsto \infty$ and $\infty \mapsto 0$, since

$$x' + \frac{u'}{\lambda - \lambda_0} = \frac{x + \lambda u}{\lambda - \lambda_0}.$$

Thus projection fibers are preserved up to a nonzero scalar. All even directions of $\mathcal{P}'$ are finite. Applying Lemma 9 again shows that both supports have exactly $j+1$ even directions, and hence

$$\mathcal{E}(\mathcal{P}) = B \cup \{\infty\}, \qquad \Lambda := \mathcal{E}(\mathcal{P}') = \{0\} \cup \{1/(\lambda - \lambda_0) : \lambda \in B\}.$$

Write the coordinates of $\mathcal{P}'$ again as (u,x) and put $\chi(y) = \prod_{\lambda \in \Lambda}(y - \lambda)$. Use $R_{\mathcal{P}'}$, $Q_{\mathcal{P}'}$, and $D_{\mathcal{P}'}$ as in Lemma 9. That lemma gives

$$D_{\mathcal{P}'} = R_{\mathcal{P}'} - Q_{\mathcal{P}'}^2 = \sum_{i=0}^{j} d_i(y) z^i, \qquad \deg d_i \le 2j + 2 - i, \qquad \deg Q_{\mathcal{P}'} \le j+1,$$

and $(y-\lambda)^2 \mid d_i(y)$ for every $\lambda \in \Lambda$. Since $\deg \chi^2 = 2j+2$, these bounds imply

$$R_{\mathcal{P}'} = Q_{\mathcal{P}'}^2 + \kappa \chi^2.$$

Here $\kappa \neq 0$, since $R_{\mathcal{P}'}$ is a product of distinct linear factors. Let $Q_{\mathcal{P}',j+1}$ be the homogeneous part of total degree $j+1$ of $Q_{\mathcal{P}'}$. Comparing highest homogeneous parts gives

$$\prod_{(u,x)\in\mathcal{P}'} (z - yu) = Q_{\mathcal{P}',j+1}(z,y)^2 + \kappa y^{2j+2}.$$

Evaluation at $(z,y) = (u_0, 1)$ for any $(u_0, x_0) \in \mathcal{P}'$ shows that $-\kappa = c^2$ for some $c \in \mathbb{F}_p^\times$. Therefore

$$R_{\mathcal{P}'} = (Q_{\mathcal{P}'} + c\chi)(Q_{\mathcal{P}'} - c\chi).$$

The factors are coprime: a common irreducible factor would divide $\chi(y)$, but no nonconstant polynomial in y divides a polynomial monic in z. They partition $\mathcal{P}'$ into disjoint sets $\mathcal{P}_1, \mathcal{P}_2$ of size $j+1$, with equal projection multisets in every direction of Λ. For a fixed point of either component, each direction supplies an opposite partner. Different directions give different partners, and there are exactly $j+1$ opposite points. Thus every cross-component line contains exactly one point of each component and has direction in Λ. Consequently

$$\mathcal{P}_1, \qquad \mathcal{P}_2, \qquad \{[1 : -\lambda : 0] : \lambda \in \Lambda\}$$

form a dual 3-net of order $j+1$. Since $j+1 < p$, Theorem 5.1 of Blokhuis, Korchmáros, and Mazzocca [2] places $\mathcal{P}_1 \cup \mathcal{P}_2$ on a conic, possibly reducible.

We use the multiplicative and additive forms of Lemma 8 of the paper.

Suppose the conic is smooth. It is defined over $\mathbb{F}_p$, since otherwise it and its Frobenius conjugate would be distinct conics sharing $2j+2 > 4$ points, contrary to Bézout's theorem. Every affine normalization below induces a projective map on directions. If the line at infinity is tangent, normalize to (t, t^2). For the component parameter sets S, T, the chord directions are $-(s+t)$, giving $|S| = |T| = |S+T| = j+1$. The additive form of Lemma 8 of the paper would give an additive subgroup of $\mathbb{F}_p$ of order $j+1$, impossible since $1 < j+1 < p$. If the two points at infinity are rational, normalize to $(t, c_0/t)$, with $c_0 \neq 0$. Chord directions are $c_0/(st)$, so $|S| = |T| = |ST| = j+1$. Hence Λ is projectively equivalent to the set of $(j+1)$st roots of unity.

If the points at infinity are nonrational, an affine normalization identifies the conic with $z^{p+1} = 1$ in $\mathbb{F}_{p^2}$. For two distinct circle points z_1, z_2,

$$(z_1 - z_2)^{p-1} = -\frac{1}{z_1 z_2}.$$

Choose $\theta \in \mathbb{F}_{p^2} \setminus \mathbb{F}_p$ with $\theta^p = -\theta$. On direction vectors,

$$(x + y\theta)^{p-1} = \frac{x - y\theta}{x + y\theta},$$

which is a nonsingular fractional linear map distinguishing the $\mathbb{F}_p$-directions. For the component parameter sets S, T in this group, the chord formula gives $|S| = |T| = |ST| = j+1$, and Lemma 8 of the paper again makes Λ projectively equivalent to the $(j+1)$st roots of unity.

For a reducible conic, a double line would give only one cross-component direction. For two distinct lines, one contains at least $j+1$ support points. Since any line meeting both components contains exactly one point of each, this line contains all of one component, and the other contains the other component. Both lines are rational, and their intersection is not a support point. For parallel lines, the normal forms $(0, s)$ and $(1, t)$ give $|S| = |T| = |S - T| = j+1$, contradicting the additive form of Lemma 8 of the paper. For intersecting lines, the forms $(s, 0)$ and $(0, t)$, with nonzero parameters, give $|S| = |T| = |TS^{-1}| = j+1$. Thus Λ is again projectively equivalent to the $(j+1)$st roots of unity.

It follows that $B \cup \{\infty\}$ is projectively equivalent to this root set. Root forms for the two sets are

$$f_1(z,y) = z^j y - y^{j+1}, \qquad f_2(z,y) = z^{j+1} - y^{j+1}.$$

Their binary Hessians are

$$\mathrm{Hess}(f_1) = -j^2 z^{j-2}\big(z^j + (j^2-1)y^j\big), \qquad \mathrm{Hess}(f_2) = -j^2(j+1)^2 z^{j-1} y^{j-1}.$$

Since $p > 2j+2$ and $j \geq 3$, they have respectively $j+1$ and two distinct projective zeros over the algebraic closure. A nonsingular linear substitution preserves the number of distinct Hessian zeros, since the Hessian transforms by that substitution and a nonzero scalar. This contradicts projective equivalence and proves (i).

Proof of (ii). Lemma 9 gives $\deg Q_{\mathcal{P}} \leq j+1$; its term z^{j+1} makes this exact. For each $\lambda \in B$, the same lemma gives $(y - \lambda)^2 \mid D_{\mathcal{P}}$. Since B is the root set of $y^j - 1$,

$$D_{\mathcal{P}} = (y^j - 1)^2 G_{\mathcal{P}}, \qquad \deg G_{\mathcal{P}} \leq 2.$$

The quotient is nonzero because $R_{\mathcal{P}}$ is not a square. Homogenize $R_{\mathcal{P}}$, $Q_{\mathcal{P}}$, and $G_{\mathcal{P}}$ to degrees $2j+2$, $j+1$, and two, respectively, by setting

$$\widetilde{R}_{\mathcal{P}}(z,y,w) = \prod_{(u,x)\in\mathcal{P}} (z-uy-xw),$$
$$\widetilde{Q}_{\mathcal{P}}(z,y,w) = w^{j+1} Q_{\mathcal{P}}(z/w, y/w),$$
$$\widetilde{G}_{\mathcal{P}}(z,y,w) = w^{2} G_{\mathcal{P}}(z/w, y/w).$$

Then

$$\widetilde{R}_{\mathcal{P}} = \widetilde{Q}_{\mathcal{P}}^2 + (y^j - w^j)^2 \widetilde{G}_{\mathcal{P}}.$$

On a support dual line $L_{u,x} : z - uy - xw = 0$, it implies

$$\left(\widetilde{Q}_{\mathcal{P}}|_{L_{u,x}}\right)^2 = -(y^j - w^j)^2 \widetilde{G}_{\mathcal{P}}|_{L_{u,x}}.$$

Unique factorization in $\mathbb{F}_p[y,w]$ shows that $-\widetilde{G}_{\mathcal{P}}|_{L_{u,x}}$ is the square of a linear form, possibly zero. If $\widetilde{G}_{\mathcal{P}}$ has rank three, every support dual line is tangent to its smooth conic, so the dual conic contains $\mathcal{P}$. If it has rank two, its two distinct linear factors over the algebraic closure restrict proportionally on each support dual line, unless the line is itself a factor. In either case, every support dual line passes through their intersection. Thus the original support is collinear, contradicting its evenness in $j \geq 3$ directions, since projection on a line is injective in all but at most one direction. In rank one, write $\widetilde{G}_{\mathcal{P}} = \alpha M^2$ with $\alpha \in \mathbb{F}_p^{\times}$ and M a nonzero linear form over $\mathbb{F}_p$. Some support dual line has $M|_{L_{u,x}} \neq 0$, and the square restriction then makes $-\alpha$ a square in $\mathbb{F}_p$. Hence $\widetilde{G}_{\mathcal{P}} = -\ell_{\mathcal{P}}^2$. Setting $w = 1$ shows that $-G_{\mathcal{P}}$ is a square in $\mathbb{F}_p[z,y]$, proving (ii).

Proof of (iii). Assume $j \geq 4$ and suppose first that $\mathcal{P}$ lies on a smooth conic. Work over an algebraic closure. An even-direction center on the line at infinity cannot lie on the conic, since projection from it is injective on the affine conic points. Each even direction therefore induces an involution of the conic preserving $\mathcal{P}$ without a support fixed point. If the line at infinity is tangent, the parametrization (t, t^2) gives involutions $t \mapsto c - t$. Two distinct ones generate a nonzero translation of order p, forcing $p \mid 2j+2$, a contradiction.

Otherwise, put the conic parameters at infinity at $0, \infty$ and use the parametrization $[t^2 : 1 : t]$. Let S be the support parameter set, whose elements are nonzero and finite. Projection involutions have the form $\sigma_c : t \mapsto c/t$, with centers $[c : -1 : 0]$.

Let Γ be the multiplicative stabilizer of S. It is a finite cyclic group, and its free action on S gives $v := |\Gamma| \mid 2j+2$. Fix a preserving reflection σ_{c_0}. Since $\sigma_c \sigma_{c_0}$ is multiplication by c/c_0, the preserving reflections have coefficients exactly in $c_0\Gamma$. There are at least j such reflections, one for each direction in B, so $v \geq j$. Together with $v \mid 2j+2$ and $j \geq 4$, this implies $v = j+1$ or $v = 2j+2$.

If $v = j+1$, then

$$S = \alpha\Gamma \cup \beta\Gamma,$$

where the two cosets are distinct. For every $c \in \alpha\beta\Gamma$, the involution $t \mapsto c/t$ exchanges these cosets. Thus it preserves S and has no fixed point in S. These give v preserving reflections, which is the total number of preserving reflections. Consequently, their coefficient set is exactly $\alpha\beta\Gamma$, and all $j+1$ of them give even directions.

If $v = 2j+2$, then $S = \alpha\Gamma$, and the preserving coefficients form $\alpha^2\Gamma$. A preserving reflection has a fixed point in S if and only if $c \in \alpha^2\Gamma^2$. Since $|\Gamma| = 2j+2$, we have $-1 \in \Gamma$ and $|\Gamma^2| = j+1$. For $c \in \alpha^2\Gamma^2$, the equation $t^2 = c$ therefore has exactly two solutions in S. The remaining $j+1$ preserving reflections have no fixed point in S and give precisely the even directions.

In both cases, the fixed-point-free coefficients form a multiplicative coset of a subgroup of order $j+1$. Hence the complete even-direction set has $j+1$ elements projectively equivalent to the $(j+1)$st roots of unity. Each center is rational, since it lies both on a chord through two distinct rational support points and on the rational line at infinity. Thus the complete even-direction set consists of B and one extra direction λ_*.

If λ_* is finite, a root form is

$$f_{\lambda_*}(z,y) = (z - \lambda_* y)(z^j - y^j) = \sum_{i=0}^{j+1} \binom{j+1}{i} c_i z^{j+1-i} y^i.$$

Its only possibly nonzero normalized coefficients are $c_0 = 1$, $c_1 = -\lambda_*/(j+1)$, $c_j = -1/(j+1)$, and $c_{j+1} = \lambda_*$. Projective equivalence to $z^{j+1} - y^{j+1}$ would express f_{λ_*} as a linear combination of two $(j+1)$st powers of linear forms. Each power contributes a rank-one matrix to $(c_{r+t})_{0\leq r\leq 2,\, 0\leq t\leq j-1}$, so this matrix would have rank at most two. Its columns $0, j-2, j-1$, however, give

$$\det \begin{pmatrix} 1 & 0 & 0 \\ -\lambda_*/(j+1) & 0 & -1/(j+1) \\ 0 & -1/(j+1) & \lambda_* \end{pmatrix} = -\frac{1}{(j+1)^2} \neq 0.$$

The binomial normalizations are valid since $p > 2j+2$. If $\lambda_* = \infty$, interchanging z, y reduces to the finite case $\lambda_* = 0$, because $B^{-1} = B$. Thus no smooth conic contains $\mathcal{P}$.

As established in the proof of (ii), $\widetilde{G}_{\mathcal{P}} = -\ell_{\mathcal{P}}^2$. Evaluating the homogeneous identity at $y = 1, w = 0$, and writing n_u for the occupancy of column u, gives

$$\prod_{u \in A} (z - u)^{n_u} = Q_{\mathcal{P},j+1}(z, 1)^2 - \ell_{\mathcal{P}}(z, 1, 0)^2,$$

where $Q_{\mathcal{P},j+1}$ is the homogeneous part of total degree $j + 1$ of $Q_{\mathcal{P}}$. If $\ell_{\mathcal{P}}(z, 1, 0)$ were zero, all n_u would be even, contrary to (i). Therefore

$$P_{\pm}(z) = Q_{\mathcal{P},j+1}(z, 1) \pm \ell_{\mathcal{P}}(z, 1, 0)$$

are distinct monic polynomials of degree $j + 1$. Their product is $\prod_{u \in A}(z - u)^{n_u}$, so all their roots lie in A, and their difference has degree at most one. This proves (iii). □

S3. Computer Certificates and Their Interpretation

The program `verify.py` generates `results.json`. The output fields and their roles in Proposition 26, Lemma 27, and Theorem 30 are described below. The computations use exact symbolic, integer, rational, and finite-field arithmetic. The command `python verify.py --check` recomputes every certificate and compares the resulting JSON object with the supplied output.

The `status` field records successful completion. The `summary` field lists the principal counts. The `proof_references` field gives the corresponding results in the paper and the supplementary PDF filename. The other fields contain the detailed calculations.

A. Algebraic identities: `symbolic_checks`

This part checks the two Hessian identities used in the proof of (i), with j left indeterminate, and the nonzero coefficient minor $-1/(j + 1)^2$ used in (iii). In the output, the latter identity is recorded as `catalectic_minor`, with $h = j + 1$.

It also verifies the square-sum estimate in Lemma 27. For $J = j + 1$ and $1 \le r \le J - 1$, put

$$E(r) = (J - r + 1)^2 + r - 1 + (r + 1)^2 + J - r - 1.$$

Then

$$J^2 + J + 2 - E(r) = 2(r - 1)(J - r - 1) \ge 0.$$

This factorization is recorded as `common_square_bound_gap` and gives $S_2 \le j^2 + 3j + 4$.

It also checks the quadratic restriction identities used in (ii). For a symmetric matrix M, restricting its quadratic form to the support dual line $z - uy - xw = 0$ gives $ay^2 + 2byw + cw^2$, and

$$b^2 - ac = -[1, -u, -x] \operatorname{adj}(M)[1, -u, -x]^{\top}.$$

Thus a square restriction in the nonsingular case places the original support point on the dual conic. For a rank-two factorization, the corresponding restriction discriminant is $(r_1 s_2 - r_2 s_1)^2$, which tests the proportionality of the restricted linear factors. These identities verify the algebraic steps in Section S2.

B. Integer and polynomial filters: `moment_screen` *and* `polynomial_screen`

These fields give the finite calculations in Theorem 30, which proves $d \ge 2j + 4$ when $1 \le k - j \le \min\{5, j - 1\}$. Put $\delta = k - j$ and $h = j + 1$. The integer multiplicity differences m_u, their evaluation polynomial $q(U)$, and the congruences for $S_v = \sum_{u \in A} m_u^v$ are those of Lemma 27. In the nonsmooth-conic branch, the two root polynomials meet the hypotheses of that lemma also when $j = 3$. The proof excludes $j \ge 6\delta$ by the sign-sum and multiplicity bounds, leaving $1 \le \delta \le 5$ and $\max\{3, \delta + 1\} \le j < 6\delta$. The program enumerates every ordered pair of partitions of a common positive sum at most h, with each part repeated at most δ times, and retains those with between h and k parts in total. Appending zeros gives every required integer value multiset. There are 69 parameter pairs and 19532 admissible multisets. For each multiset the program also checks $0 < S_2 \le j^2 + 3j + 4$ and $2 \mid S_2$, as in Lemma 27.

For each multiset, it checks all required moment congruences and retains primes satisfying $p > 2k$ and $\operatorname{lcm}(j, k) \mid p - 1$. The field `moment_survivors` lists the 43 surviving quadruples consisting of p, j, k and the integer multiset. They occur in exactly three cases:

(p, j, k)	target multisets	polynomials tested	matches
$(13, 4, 6)$	2	13	$\pm 2(U^2 - U)$
$(19, 6, 9)$	5	361	none
$(31, 10, 15)$	36	923521	none.

All target multisets contain zero, as recorded by `all_targets_contain_zero`. If $q(u_0) = 0$ for $u_0 \in A$, then replacing $q(U)$ by $q(u_0 U)$ permutes its evaluation multiset and gives $q(1) = 0$. Consequently the enumeration of $q(U) = \sum_{i=2}^{\delta} c_i(U^i - U)$ is exhaustive after normalization. The field `matching_coefficients_q1_to_qdelta` lists coefficients modulo p in increasing degree order. The two lists $[11, 2]$ and $[2, 11]$ in characteristic 13 are precisely $q(U) = \pm 2(U^2 - U)$. Therefore only the first parameter triple requires the boundary-code check below.

C. Complete boundary-code enumeration: `boundary_code`

For $(p, j, k) = (13, 4, 6)$, the program reconstructs the original binary parity-check matrix using $x + \lambda u$, where A and B are the multiplicative subgroups of orders k and j. Gaussian elimination gives length 78, rank 49, and kernel dimension 29. Every computed basis vector is checked against every original parity equation, and its free coordinate ensures independence from the other basis vectors.

The field `kernel_basis_hex` stores these 29 binary vectors as hexadecimal integers. Bit $tp + x$ corresponds to (a^t, x), with $a = g^{(p-1)/k}$ and g the least positive primitive root selected by the program. Thus bit zero is the coordinate $(1, 0)$. The program enumerates all $2^{29} = 536870912$ binary combinations of this basis. Its field `weight_enumerator` maps each occurring weight to its number of words; these counts sum to 2^{29}, and the least positive weight with nonzero count is 16. The field `weight16_witness_hex` supplies an attaining word whose syndrome is checked directly. Consequently the boundary code has $d = 16$, and a support of size $2j + 2 = 10$ is impossible. The hexadecimal representation is only a compact encoding of the exact binary vectors.

D. Exceptional coefficient: `coefficient_check`

For the smooth-conic branch with $j = 3$, $k = 5$, write w for a primitive fifth root and consider the coefficient of X^5 in $(1 - X^5)((1 - X)(1 - wX))^{-1/2}$. The output field `reduced_coefficient_w0_to_w3` records its coefficients modulo $\Phi_5(w)$ in increasing powers of w:

$$\left[-\frac{165}{256},\ 0,\ -\frac{5}{256},\ -\frac{5}{256}\right].$$

Writing $s = w^2 + w^3$, the relation $s^2 + s - 1 = 0$ shows that vanishing requires $s = -33$, whence $1055 = 5 \cdot 211 = 0$ in the field. Since the admissible prime is greater than 10, only $p = 211$ remains. The output records this exact obstruction, its prime factors, and the unique admissible prime.

E. Tree certificate: `p211_ball`

At $(p, j, k) = (211, 3, 5)$, the program expands the Tanner graph about $(1, 0)$ using

$$(u, x) \sim (\lambda, x + \lambda u), \qquad u \in A, \quad \lambda \in B.$$

The fields `A` and `B` give the actual subgroup labels. In `nodes`, the tags `v` and `c` denote variable and check nodes; each record contains the node label, its distance from the root, and its parent. The layer sizes through radius five are $1, 3, 12, 24, 96, 192$, and `non_tree_edges` is zero. Edges incident with every vertex at distance less than five are checked; bipartiteness excludes edges between two vertices in the last layer.

The maps $(u, x) \mapsto (cu, cx + t)$ and $(\lambda, z) \mapsto (\lambda, cz + t)$, with $c \in A$ and $t \in \mathbb{F}_p$, preserve adjacency and act transitively on variable nodes. Any cycle of length at most ten could therefore be moved through the root and would lie in its radius-five ball, contradicting the tree certificate. Hence the girth is at least twelve, and Proposition 16 of the paper gives $d \geq 14$, excluding weight eight in this final branch.

The integer and polynomial filters, complete boundary-code enumeration, and exceptional-prime tree certificate complete the finite calculations in Theorem 30. The accompanying `README.md` gives the installation and execution commands.

REFERENCES

[1] L. Rédei, *Lacunary Polynomials Over Finite Fields*. Amsterdam, The Netherlands: North-Holland, 1973.

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